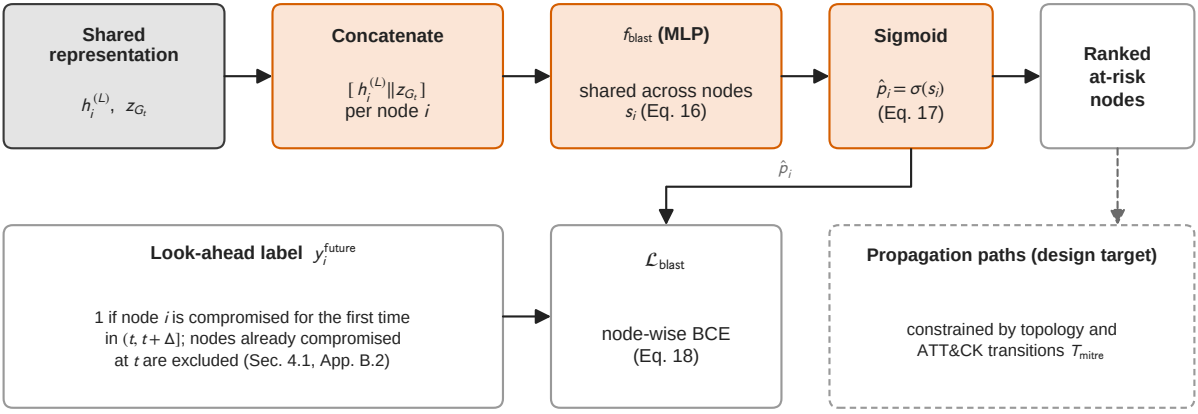Implemented and jointly trained; not evaluated in this paper (Sec. 5.4).